WEB Development Course

Project - Part A

Subject: Python Learning Platform - PyLearn

Group 21

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1 General Overview

This web application is an interactive platform designed to teach Python programming to beginners and early-stage university students.

The platform is specifically aimed at students who struggle to keep up with the fast pace of academic programming courses. It provides a structured learning path composed of progressive stages, where each stage focuses on a specific programming concept.

The system combines learning and practice through coding exercises, including both free code writing and code completion tasks. The platform also incorporates gamification elements such as experience points (XP), levels, and daily streaks in order to maintain user motivation and engagement.

The application includes three main pages:

- Learning Path Page (/path) - Displays the progression map of stages.
- Login Page (/login) - Allows users to authenticate into the system.
- Exercise Page (/exercise) - Allows users to solve coding tasks and receive feedback.

The system processes user input on the server side, evaluates code correctness, updates user progress, and returns feedback to the user.

2 UX Requirements

a. Who is the user?

The target users are:

- University students enrolled in programming courses
- Beginners with no prior programming experience
- Users with basic knowledge who want to strengthen their skills

b. What are the user's needs?

- A structured and gradual learning process
- Practical coding experience rather than theoretical learning
- The ability to learn at their own pace
- Continuous motivation to keep learning
- Immediate feedback on performance

c. What service does the application provide?

The application provides an interactive Python learning environment that includes:

- A structured learning path divided into stages
- Coding exercises (free coding and code completion)
- Real-time feedback on submitted solutions
- A progression system based on XP and levels
- Motivation mechanisms such as daily streak tracking

Overall, the system aims to combine learning, practice, and motivation in a single platform.

d. What is the exact process that enables the service?

1. The user logs into the system through the login page.
2. The user is redirected to the learning path page.
3. The user selects a stage.
4. The system presents a task that includes a short explanation, instructions, and a coding area.
5. The user submits their solution.
6. The user input is sent from the client to the server.
7. The server performs code validation, correctness checking, XP calculation, and progress updates.
8. The server sends a response back to the client, including correct/incorrect feedback, an optional solution, and updated progress status.
9. The user continues to the next stage.

e. What content is required from the user?

The application requires the following inputs:

- Email address
- Password
- Code input written by the user
- Answers for code completion tasks

Additionally, the system stores:

- User progress
- XP points
- Daily streak data

3 UI Specifications

a. What impression should the application create?

- Playfulness (gamification)
- Simplicity and accessibility
- Motivation and progress

b. UI Specification Document

Document Info

- Project: PyLearn
- Authors: Elad Atias, Romi Ben Shahar and Hadar Ben Ami
- Date: 30/03

1. General Overview

This application is a web-based Python learning platform designed to help users improve their coding skills through interactive exercises and a structured learning path. The system includes three main pages:

1. Learning Path Page (/path) - Displays the user's progress, levels, and stages in a structured and gamified path.
2. Login Page (/login) - A simple form that enables users to log into their account.
3. Exercise Page (/exercise) - Allows users to practice Python by solving coding exercises and receiving immediate feedback.

The interface will be responsive, clean, and engaging, focusing on simplicity, accessibility, and motivation through gamification elements.

2. Learning Path Page (/path)

Purpose - The learning path page presents the user's progression through different stages and encourages continued engagement.

Components:

- **Header**
  - Contains the platform logo on the left.
  - Navigation menu on the right with links to Path, Profile, and Logout.
- **Progress Section**
  - Displays the user's current level and XP.
  - Shows a progress bar indicating advancement.
- **Stages Map**
  - Visual representation of the stages in a path-based layout.
  - Each stage is represented by an icon.
  - Locked and unlocked states for stages.
  - Completed stages are visually marked.
- **Footer**
  - Basic navigation links and legal links.

3. Exercise Page (/exercise)

Purpose - To allow users to practice Python by solving coding exercises and receiving feedback.

Components:

- **Header**
  - Same as the learning path page header.
- **Task Section**
  - Title of the exercise.
  - Short explanation of the concept.
  - Task instructions.
- **Code Editor**
  - Text area for writing Python code.
  - Pre-filled template for code completion tasks when needed.
- **Submit Button**
  - Sends the user input to the server.
- **Feedback Area**
  - Displays whether the answer is correct or incorrect.
  - Shows hints or a full solution if necessary.
- **Navigation Controls**
  - Button to move to the next stage.

4. Login Page (/login)

Purpose - To allow users to log into their account.

Components:

- **Login Form (Centered)**
  - Email address (required).
  - Password (required).
- **Submit Button**
  - Validates input and authenticates the user.
- **Error Messages**
  - Displayed for invalid or missing credentials.

5. UI Behavior Specifications

Input Validation:

- All required fields must be validated before submission.
- Code submissions must not be empty.

Error Messages:

- Displayed clearly near the relevant field or section.

Responsive Design:

- The layout adapts to both desktop and mobile devices.

Animations and Interactions:

- Visual feedback on correct answers.
- Smooth transitions between stages.

- Hover effects on buttons.

6. Visual Design Notes

Color Scheme:

Primary Color	#58CC02 - Bright Green
Accent Color	#1CB0F6 - Light Blue
Background	#F7F7F7 - Light Gray-White

Typography:

- Font family: Open Sans, sans-serif
- Font sizes: 16px for body text, 24px for section titles, 36px for the main page headline.